University of Aberdeen
Applied Statistics - PU5017
Assessment 1 Report

Fernaldy Aristo Wirjowerdojo
52642998

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Task 1

Summarise the data for all subjects overall and separately for those who did and did not have coronary artery disease (CAD)

Some initial checks show that the variables *Total cholesterol at baseline, Total cholesterol after follow-up, and Low-density lipoprotein* are normally distributed while *Triglycerides and Thyroid-stimulating hormone* are not normally distributed.

Variables	**Overall (n = 7932)**	**CAD (n = 2125)**	**No CAD (n = 5807)**
Total cholesterol at baseline (mg/dL), *mean (SD)*	231.67 (37.06)	240.78 (36.87)	228.34 (36.57)
Total cholesterol at follow-up (mg/dL), *mean (SD)*	191.26 (39.96)	200.42 (36.59)	187.91 (36.53)
Low-density lipoprotein (mg/dL), *mean (SD)*	131.70 (19.38)	136.03 (18.91)	130.12 (19.30)
Triglycerides (mg/dL), *median [IQR]*	140.00 [99.00, 198.00]	149.00 [105.00, 218.00]	136.00 [97.00, 192.00]
Thyroid-stimulating hormone (mg/dL), *median [IQR]*	1.64 [1.21, 2.10]	1.87 [1.33, 2.18]	1.55 [1.18, 2.06]

Table 1: Summary statistics of continuous variables

A total of 7,932 participants were included in the analysis (2,125 with CAD and 5,807 without). Those with CAD showed overall higher levels in all the variables: total cholesterol at baseline and at follow-up, low-density lipopretein, triglycerides, and thyroid-stimulating hormone compared to those without CAD.

Variable	**Overall** ($n = 7,932$)	**CAD** ($n = 2,125$)	**No CAD** ($n = 5,807$)
Gender, n (%)			
Female	4,009 (50.5)	798 (37.6)	3,211 (55.3)
Male	3,932 (49.5)	1,327 (62.4)	2,596 (44.7)
Level of Physical Activity, n (%)			
No activity	2,402 (30.3)	761 (35.8)	1,641 (28.3)
Moderate activity	3,556 (44.8)	965 (45.4)	2,591 (44.6)
Strenuous activity	1,974 (24.9)	399 (18.8)	1,575 (27.1)
Congenital Anomalies, n (%)			
Yes	4 (14.8)	1 (25.0)	3 (13.0)
No	23 (85.2)	3 (75.0)	20 (87.0)
BMI Category, n (%)			
Underweight & Normal weight	3,136 (39.5)	853 (40.1)	2,283 (39.3)
Overweight	2,824 (35.6)	677 (31.9)	2,147 (37.0)
Obese	1,972 (24.9)	595 (28.0)	1,377 (23.7)
Smoking Status, n (%)			
Non-smoker	6,043 (76.2)	1,317 (62.0)	4,726 (81.4)
Smoker	1,889 (23.8)	1,808 (38.0)	1,081 (18.6)

Table 2: Frequency charts of categorical variables

The study size is fairly balanced between male and female with 49.5% male and 50.5% female. The overall participants' level of physical activity is lopsided in the middle (moderate activity) with around 44.8% of them. Congenital anomalies were mostly unreported (missing). 24.9% of the participants were obese and 35.6% of them were also overweight. Furthermore, a majority of the participants were non-smoker (76.2%).

Task 2

a. Does Thyroid-stimulating hormone differ between male and female participants?

Thyroid-stimulating hormone (TSH) is a continuous variable while ($gender$) is binary. So, the independent t-test will be used and the assumptions for that test will be checked first. The following hypothesis is considered:

- H_0: There is no difference in the mean thyroid-stimulating hormone levels between male and female participants in the population
- H_0: There is a difference in the mean thyroid-stimulating hormone levels between male and female participants in the population

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Thyroid-stimulating hormone, mg/dL	female	4003	1.21	.294	.005
	male	3923	2.10	.298	.005

Figure 1: Summary statistics of TSH between male and female

The summary statistics show that the mean TSH levels are higher in male than female, but the significance needs to be checked.

The assumptions to check are:

- Independent groups
- TSH levels are normally distributed in both groups
- Variance of the TSH levels are equal between both groups

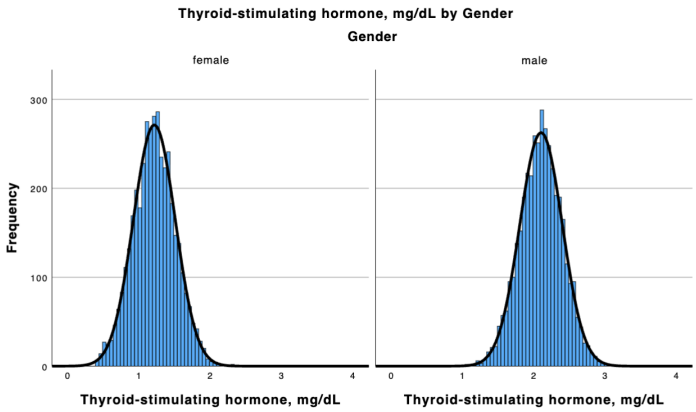


Figure 2: Distribution of TSH between male and female

Homogeneity of Variance Test

		Levene's Test for Equality of Variances	
		F	Sig.
Thyroid-stimulating hormone, mg/dL	Equal variances assumed	.307	.579

Figure 3: Levene's test for equality of variance of TSH

All the assumptions are met, both groups are independent of one another, the TSH variable is normally distributed in both groups, and the Levene's test ($p = 0.579 > 0.05$) indicates that the variance is homogeneous.

Independent Samples Test

		t-test for Equality of Means							
				Significance				95% Confidence Interval of the Difference	
		t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Thyroid-stimulating hormone, mg/dL	Equal variances assumed	-133.151	7924	<.001	<.001	-.886	.007	-.899	-.873
	Equal variances not assumed	-133.135	7915.677	<.001	<.001	-.886	.007	-.899	-.873

Figure 4: Independent t-test result

Since the p-value is significant ($p < 0.001$), the H_0 is rejected. There is a difference in the mean thyroid-stimulating hormone levels between male and female participants in the population.

In conclusion, female participants had lower thyroid-stimulating hormone levels with an average difference of 0.886 mg/dL. Furthermore, the 95% confidence interval $(-0.899, -0.873)$ indicates that mean thyroid-stimulating hormone levels in the female population is between 0.873 and 0.899 lower than that of the male population.

b. Does the level of Triglycerides differ among different levels of physical activity?

Triglycerides (*triglycerides*) is a continuous variable while physical activity (*physical_activity*) is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the distribution of triglycerides across all physical activity groups in the population
- H_1: There is a difference in the distribution of triglycerides across all physical activity groups in the population

Since there are 3 groups, the test that should be done is ANOVA. So the following three assumptions are checked first.

- The three physical activity groups are independent
- The distribution of *triglycerides* is normal in the three groups

- The variance of *triglycerides* is equal between the three groups

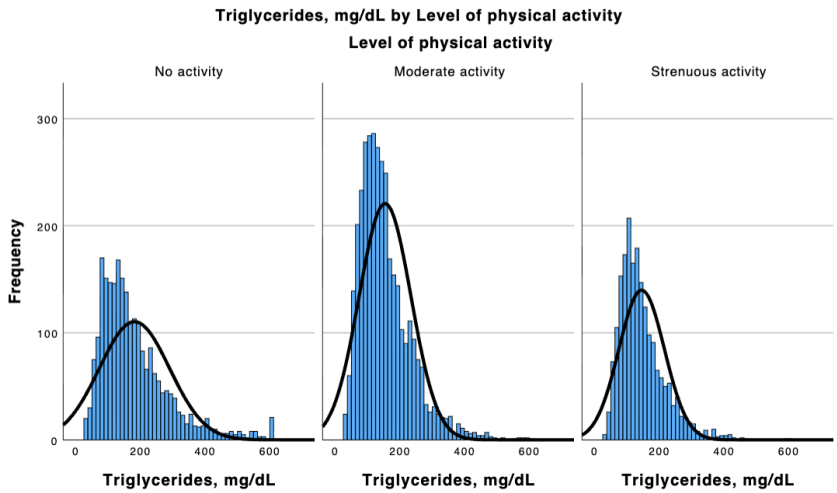


Figure 5: Distribution of triglycerides between the three physical activity groups

The first assumption is met but the figure above shows that the second assumption fails because the distribution of triglycerides for all the three groups are not normal and instead are right-skewed. Instead of the regular ANOVA, the Kruskal-Wallis test is used and the hypothesis is adjusted for non-parametric. Before the test results, the median and the IQR can be considered.

Activity Levels	**Median**	**Lower Quartile**	**Upper Quartile**
No Activity	153.50	104.00	228.00
Moderate Activity	137.50	97.00	192.00
Strenuous Activity	130.00	97.00	178.00

Table 3: Median and IQR of triglycerides levels across physical activity

The median from the three groups show that the triglycerides levels are generally lower as the level of physical activity gets more strenuous. But the Kruskal-Wallis test will be used to determine its significance.

Hypothesis Test Summary

	Null Hypothesis	Test	Sig.[a,b]	Decision
1	The distribution of Triglycerides, mg/dL is the same across categories of Level of physical activity.	Independent-Samples Kruskal-Wallis Test	<.001	Reject the null hypothesis.

a. The significance level is .050.

b. Asymptotic significance is displayed.

Figure 6: Kruskal-Wallis test result

The test result shows that the difference is significant ($p < 0.001$). The null hypothesis is rejected.

In conclusion, the distribution of triglycerides is different across the physical activity groups of no activity, moderate activity, and strenuous activity in the population. The triglycerides levels are lower as more intense the physical activity is.

c. Is there an association between physical activity and gender?

Physical activity (*physical_activity*) is an ordinal variable and gender (*gender*) is a binary variable. So, the Chi-square test for trend will be used to determine whether there is an association between physical activity and gender. The following are the null and alternate hypothesis:

- H_0: There is no association between physical activity and gender in the population
- H_1: There is an association between physical activity and gender in the population

Gender * Level of physical activity Crosstabulation

			Level of physical activity			
			No activity	Moderate activity	Strenuous activity	Total
Gender	female	Count	1260	1768	981	4009
		% within Gender	31.4%	44.1%	24.5%	100.0%
	male	Count	1142	1788	993	3923
		% within Gender	29.1%	45.6%	25.3%	100.0%
Total		Count	2402	3556	1974	7932
		% within Gender	30.3%	44.8%	24.9%	100.0%

Figure 7: Crosstabulation of gender and physical activity

The figure above show a similar trend in the level of physical activity between male and female where the majority have moderate level of physical activity

with 44.1% and 45.6% on female and male respectively, followed with 31.4% and 29.1% with no activity, and 24.% and 25.3% with strenuous activity.

However, the assumptions will be checked first:

- No more than 20% of the cells have an expected count < 5
- No cell has an expected count < 1

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	5.050[a]	2	.080
Likelihood Ratio	5.052	2	.080
Linear-by-Linear Association	3.610	1	.057
N of Valid Cases	7932		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 976.30.

Figure 8: Chi-square test result of gender and physical activity

All the assumptions are met and since the p-value from the linear-by-linear association show that it is not significant ($p = 0.057 \geq 0.05$), the null hypothesis is not rejected. **Thus**, there is no association between gender and physical activity in the population.

d. Does low-density lipoprotein differ among different BMI categories?

Low-density lipoprotein (LDL) is a continuous variable while BMI categories (BMI_cat) is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population
- H_1: There is a difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population

Since this is a continuous against a categorical variable, ANOVA will be used to determine the significance. But, the assumptions will be checked first for ANOVA:

- The BMI categories are independent
- The distribution of ldl is normal for the three BMI categories
- The variance of ldl is equal for the three BMI categories

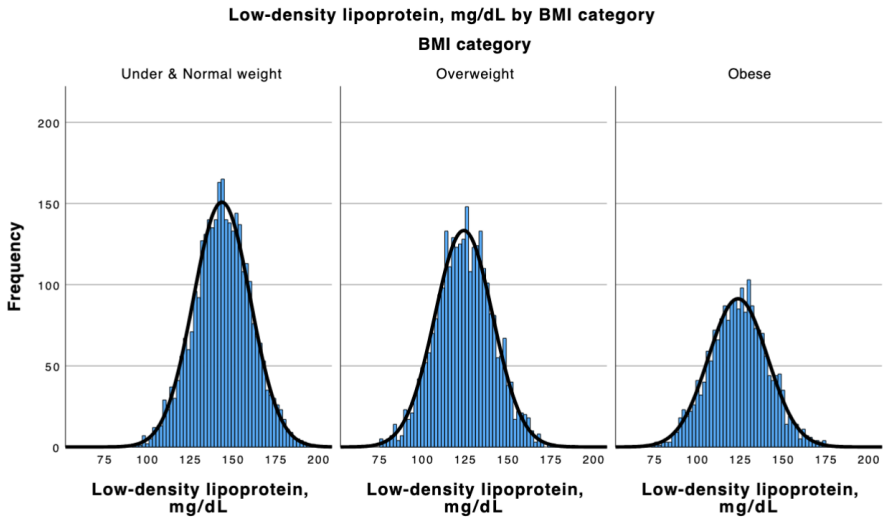


Figure 9: Distribution of ldl between the three BMI categories

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.[a]
Low-density lipoprotein, mg/dL	Based on Mean	1.479	2	7929	.228
	Based on Median	1.456	2	7929	.233
	Based on Median and with adjusted df	1.456	2	7921.631	.233
	Based on trimmed mean	1.485	2	7929	.227

a. Confidence Interval: 95%

Figure 10: Levene's test for equality of variance of ldl

The first assumption is immediately met. The two figures above show that the distribution of LDL across the three BMI categories are normal and the result of Levene's test of homogeneity of variance ($p = 0.228 > 0.05$) indicates that the variance is equal across the three BMI categories. All three assumptions for ANOVA is met.

ANOVA

Low-density lipoprotein, mg/dL

	Sum of Squares	df	Mean Square	F	Sig.[a]
Between Groups	722980.139	2	361490.069	1271.414	<.001
Within Groups	2254382.852	7929	284.321		
Total	2977362.991	7931			

a. Confidence Interval: 95%

Figure 11: ANOVA results

Multiple Comparisons

Dependent Variable: Low-density lipoprotein, mg/dL

Scheffe

(I) BMI category	(J) BMI category	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Under & Normal weight	Overweight	19.322*	.437	<.001	18.25	20.39
	Obese	19.810*	.485	<.001	18.62	21.00
Overweight	Under & Normal weight	-19.322*	.437	<.001	-20.39	-18.25
	Obese	.488	.495	.615	-.72	1.70
Obese	Under & Normal weight	-19.810*	.485	<.001	-21.00	-18.62
	Overweight	-.488	.495	.615	-1.70	.72

*. The mean difference is significant at the 0.05 level.

Figure 12: Post-hoc Scheffe test

In conclusion, The ANOVA test ($p < 0.001$) indicates that the H_0 is rejected. Thus, there is a difference in the mean low-density lipoprotein across the three BMI categories under & normal weight, overweight, and obese in the population. Furthermore, the post-hoc test shows where the significant difference in the mean LDL between the groups are:

- Under & normal weight is around 19.3 mg/dL higher than overweight
- Under & normal weight is around 19.8 mg/dL higher than obese

While the difference between the overweight and obese groups is not statistically significant.

e. Is there an association between smoking and CAD?

Both variables smoking (*smoking*) and CAD (*CAD*) are binary, so the continuity corrected chi-square test will be used with the following hypothesis:

- H_0: There is no association between smoking and coronary artery disease (CAD) status in the population
- H_1: There is an association between smoking and coronary artery disease (CAD) status in the population

Smoking Status * Coronary artery disease (CAD) status Crosstabulation

			Coronary artery disease (CAD) status		
			No CAD	CAD	Total
Smoking Status	Non-smoker	Count	4726	1317	6043
		% within Smoking Status	78.2%	21.8%	100.0%
	Smoker	Count	1081	808	1889
		% within Smoking Status	57.2%	42.8%	100.0%
Total		Count	5807	2125	7932
		% within Smoking Status	73.2%	26.8%	100.0%

Figure 13: Crosstabulation between smoking and CAD

The figure above shows that only a portion of non-smokers (21.8%) don't have coronary artery disease while almost half the smokers (42.8%) have coronary artery disease.
However the assumptions need to be checked first:

- All four cells have expected values > 5
- The two groups are independent

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	322.978[a]	1	<.001		
Continuity Correction[b]	321.910	1	<.001		
Likelihood Ratio	303.903	1	<.001		
Fisher's Exact Test				<.001	<.001
Linear-by-Linear Association	322.938	1	<.001		
N of Valid Cases	7932				

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 506.07.
b. Computed only for a 2x2 table

Figure 14: Chi-square result between smoking and CAD

All the assumptions are met and the p-value shows that it is significant ($p < 0.001$), so the H_0 is rejected. **Thus**, there is an association between smoking and coronary artery disease status in the population.

f. Is there an association between congenital anomalies and CAD?

Both variables congenital anomalies ($cong_ano$) and CAD (CAD) are binary, so the continuity corrected chi-square test will be used if the assumptions are met, otherwise the Fisher's exact test will be used instead with the following hypothesis:

- H_0: There is no association between congentional anomalies and coronary artery disease (CAD) status in the population
- H_1: There is an association between congentional anomalies and coronary artery disease (CAD) status in the population

Congenital anomalies * Coronary artery disease (CAD) status Crosstabulation

			Coronary artery disease (CAD) status		
			No CAD	CAD	Total
Congenital anomalies	Yes	Count	3	1	4
		% within Congenital anomalies	75.0%	25.0%	100.0%
	No	Count	20	3	23
		% within Congenital anomalies	87.0%	13.0%	100.0%
Total		Count	23	4	27
		% within Congenital anomalies	85.2%	14.8%	100.0%

Figure 15: Crosstabulation between smoking and CAD

The figure above shows that only a portion of people without congenital anomalies (13.0%) have coronary artery disease while people with congenital anomalies have a higher likelihood of also having coronary artery disease (25.0%). However the assumptions need to be checked first:

- All four cells have expected values > 5
- The two groups are independent

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	.386[a]	1	.534		
Continuity Correction[b]	.000	1	1.000		
Likelihood Ratio	.342	1	.559		
Fisher's Exact Test				.495	.495
Linear-by-Linear Association	.372	1	.542		
N of Valid Cases	27				

a. 3 cells (75.0%) have expected count less than 5. The minimum expected count is .59.

b. Computed only for a 2x2 table

Figure 16: Chi-square result between smoking and CAD

The first assumption is not met, so the Fisher's exact test will be used. The p-value shows that it is not significant ($p = 0.495 \geq 0.05$), so the H_0 is not rejected. **Thus**, there is no association between congenital anomalies and coronary artery disease status in the population.

g. Is there a change in Total cholesterol between baseline and after intervention?

Total cholesterol base and after (*Tchol_base* and *Tchol_after*) are continuous variables and since it wants to know if there is a change between baseline and after, the paired t-test will be considered and the assumption *The difference between total cholesterol at baseline and after intervention is normally distributed* will be checked first and the following hypothesis will be used:

- H_0: The mean total cholesterol between baseline and after intervention is the same in the population
- H_1: The mean total cholesterol between baseline and after intervention is not the same in the population

The new variable *diff_tchol* is introduced by subtracting *Tchol_base* from *Tchol_after*

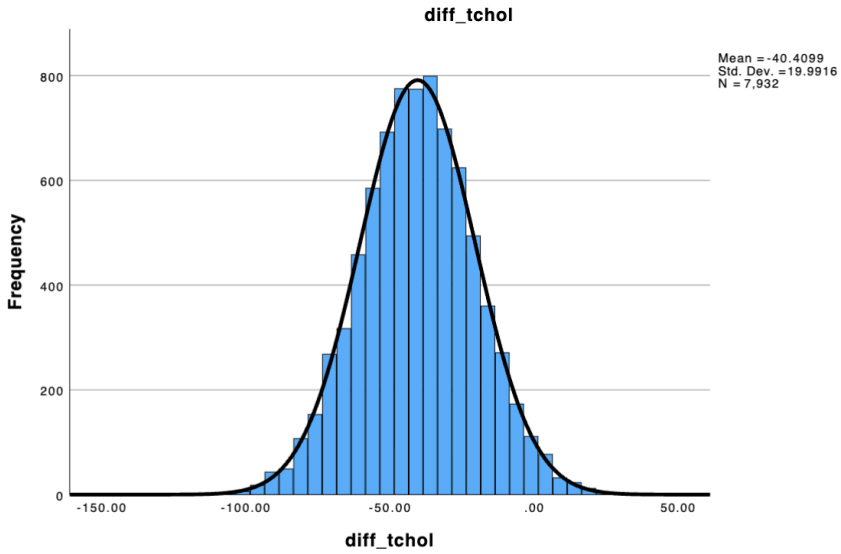


Figure 17: Difference in total cholesterol at baseline and after intervention

The figure above shows the new variable is normally distributed. So, the assumption is met.

Paired Samples Test

		Paired Differences					t	df	Significance	
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			One-Sided p	Two-Sided p
Pair 1	Total cholesterol at baseline, mg/dL - Total cholesterol after follow-up, mg/dL	40.410	19.992	.224	39.970	40.850	180.024	7931	<.001	<.001

Figure 18: Paired t-test

The t-test result ($p < 0.001$) shows that it is significant and that the H_0 should be rejected.

In conclusion, the mean total cholesterol between baseline and after intervention is different in the population. The average total cholesterol after intervention is around 40 mg/dL lower. Whereas the 95% confidence interval for the mean difference is $(39.97, 40.85)$ indicating that the mean total cholesterol after intervention in the population is between 39.97 mg/dL lower or 40.85 mg/dL higher.